

HEI Shift Light

01 - Root

02 - Input Power

Input Power

File: input_power.kicad_sch

03 - 5V Power Supply

5V Power Supply

File: 5v_power_supply.kicad_sch

04 - Input Signal

Input Signal

File: input_signal.kicad_sch

05 - Microcontroller

Microcontroller

File: microcontroller.kicad_sch

06 - USB UART Bridge

USB UART Bridge

File: USB_UART_Bridge.kicad_sch

07 - RGB LED

RGB LED

File: rgb_led.kicad_sch

08 - Mechanical

Mechanical

File: mechanical.kicad_sch

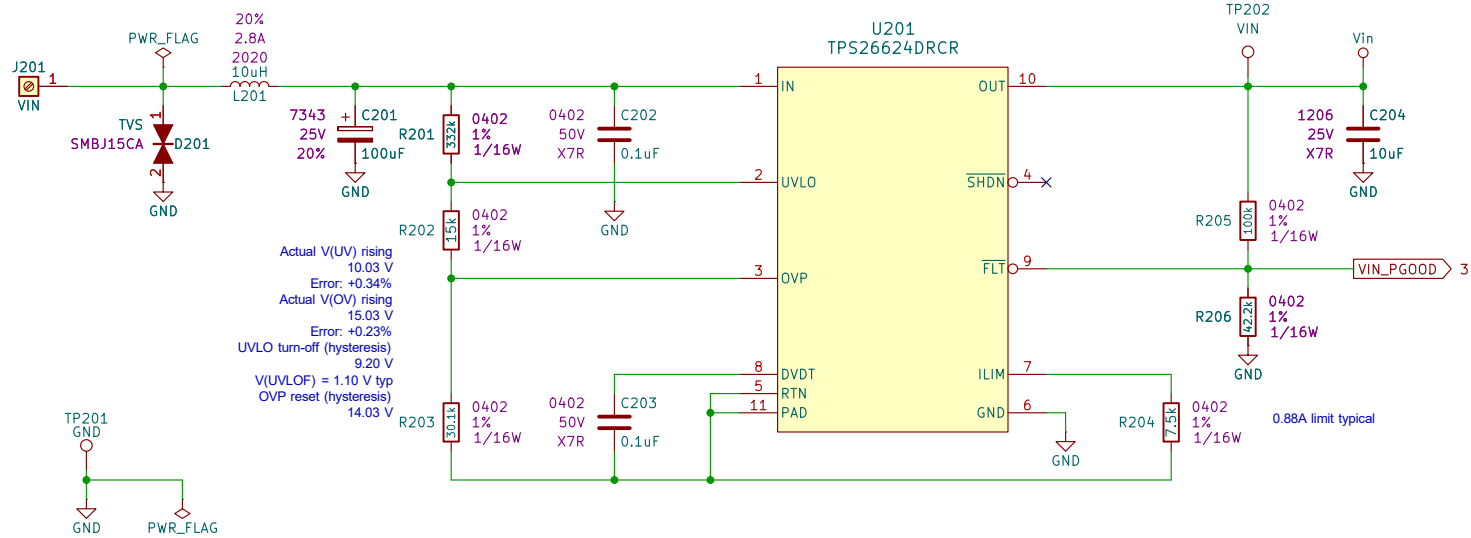
Drew Maatman

Sheet: /
File: HEI_Shift_Light.kicad_sch

Title: HEI Shift Light

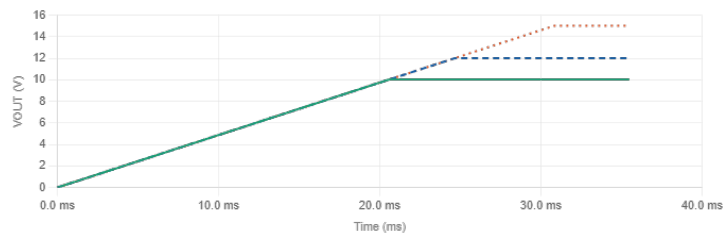
Size: A Date: 2026-06-02
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DVDT cap:
Min corner: I=2.33 μ A, GAIN=23.9 V/V (longest ramp)
Typ: I=1.98 μ A, GAIN=24.6 V/V
Max corner: I=1.68 μ A, GAIN=25.2 V/V (shortest ramp - slowest I, widest gain)

VOUT target Rise time - min Rise time - typ Rise time - max Slew rate (typ)
10.03 V (UVLO threshold) 18.0 ms 20.6 ms 23.7 ms 0.49 V/ms
12 V (Nominal) 21.5 ms 24.6 ms 28.3 ms 0.49 V/ms
15.03 V (OVP threshold) 27.0 ms 30.9 ms 35.5 ms 0.49 V/ms



VIN_PGOOD divider:
VEN at VUV (min VOUT)
2.98 V
Must be ≥ 0.65 V ≤ 0.65 V
VEN at VOV (max VOUT)
4.46 V
 ≤ 0.65 V Within safe range
Divider current at VUV
70.53 μ A
Keep ≤ 10 μ A EN leakage
 ≤ 0.65 V(EN) min = 2.98 V (need ≥ 0.65 V) ≤ 0.65 V(EN) max = 4.46 V ≤ 0.65 V Fault: FLT pulls EN to ~ 0 V (< 0.3 V)

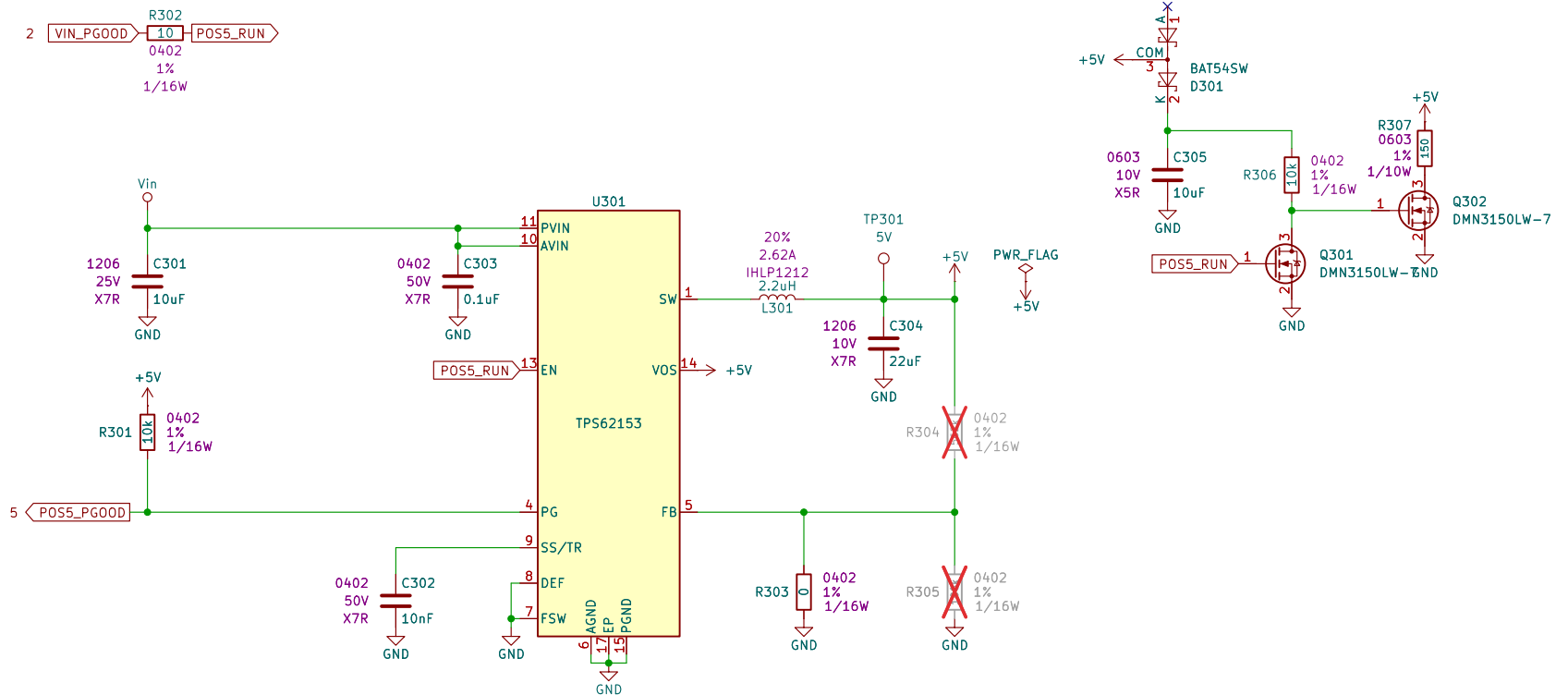
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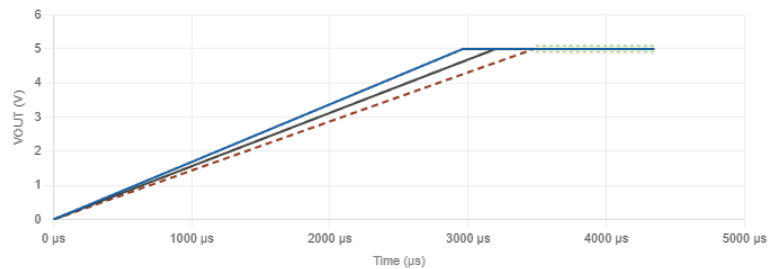
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Rise times & settled output range

Min time: I=2.7 μ A (fastest) | Typ: I=2.5 μ A | Max time: I=2.3 μ A (slowest)

Condition Rise time - min Rise time - typ Rise time - max Settled VOUT
 DEF=low (nominal 5.00 V) 2.963 ms 3.200 ms 3.478 ms 4.910 - 5.090 V
 DEF=high (+5% = 5.25 V) 2.963 ms 3.200 ms 3.478 ms 5.155 - 5.345 V



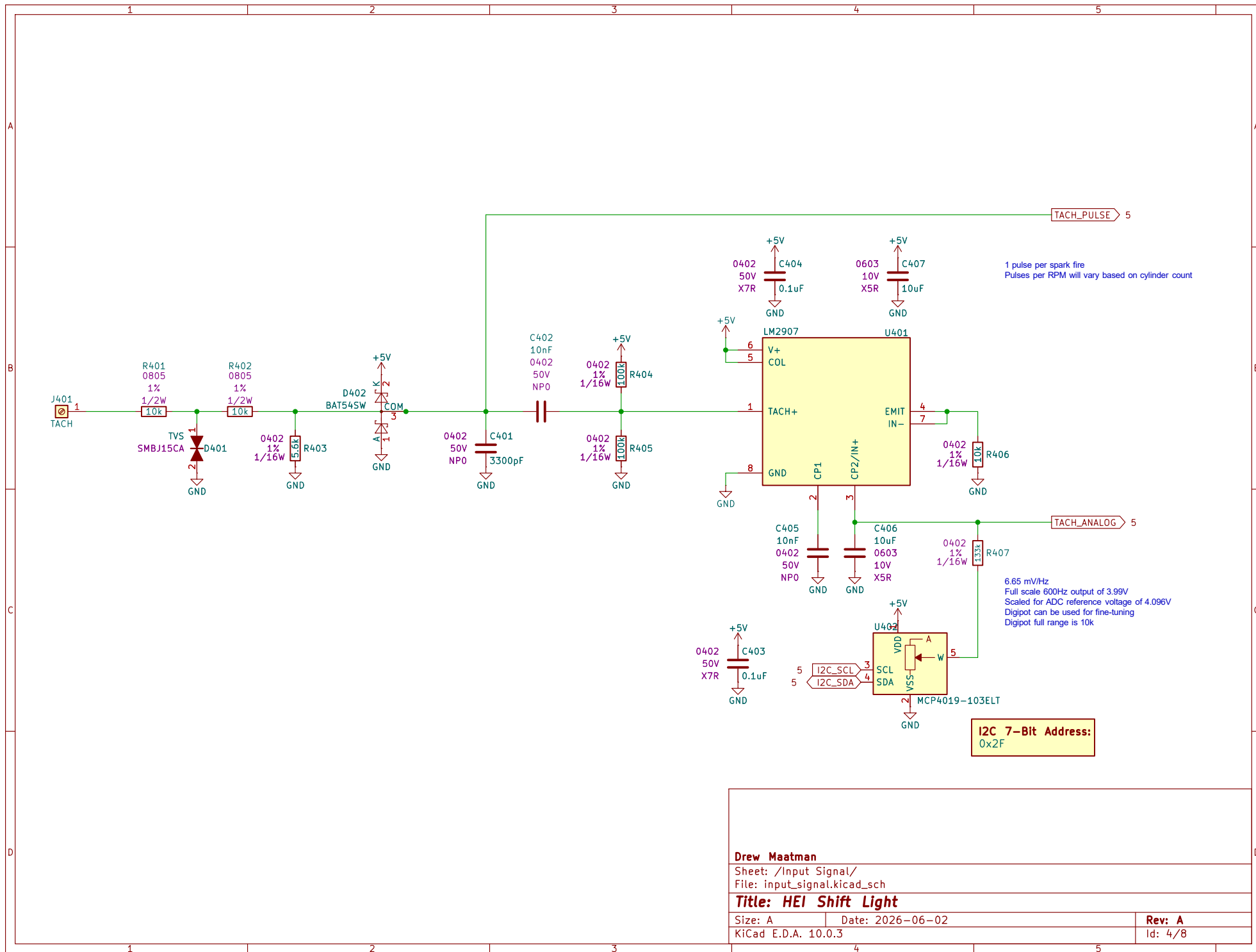
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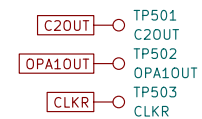
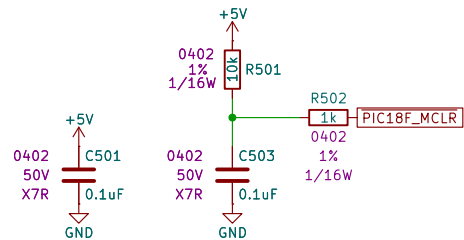
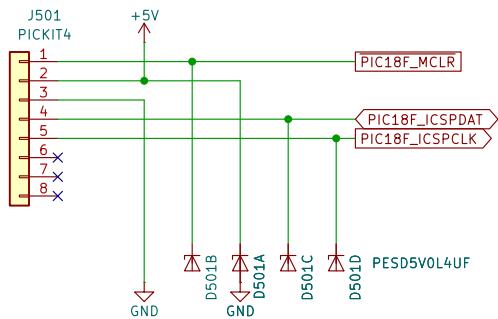
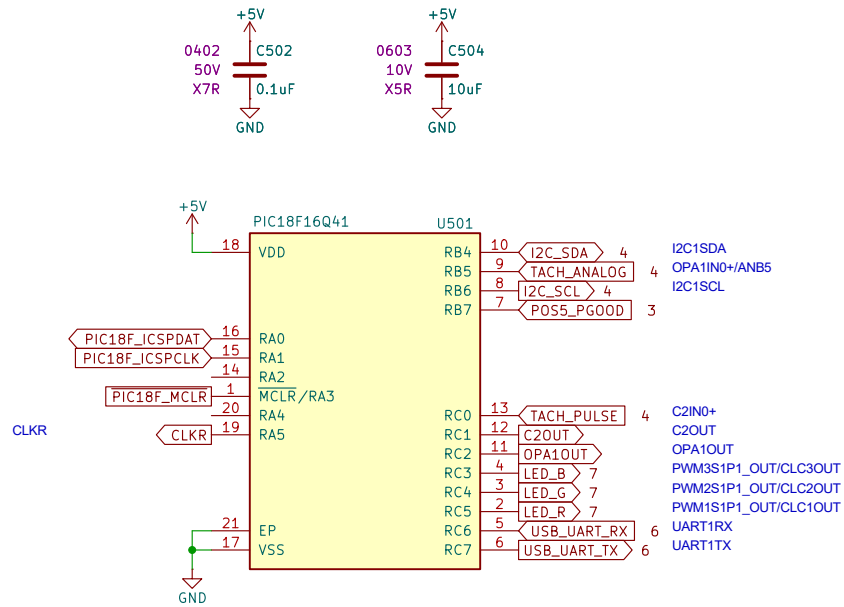
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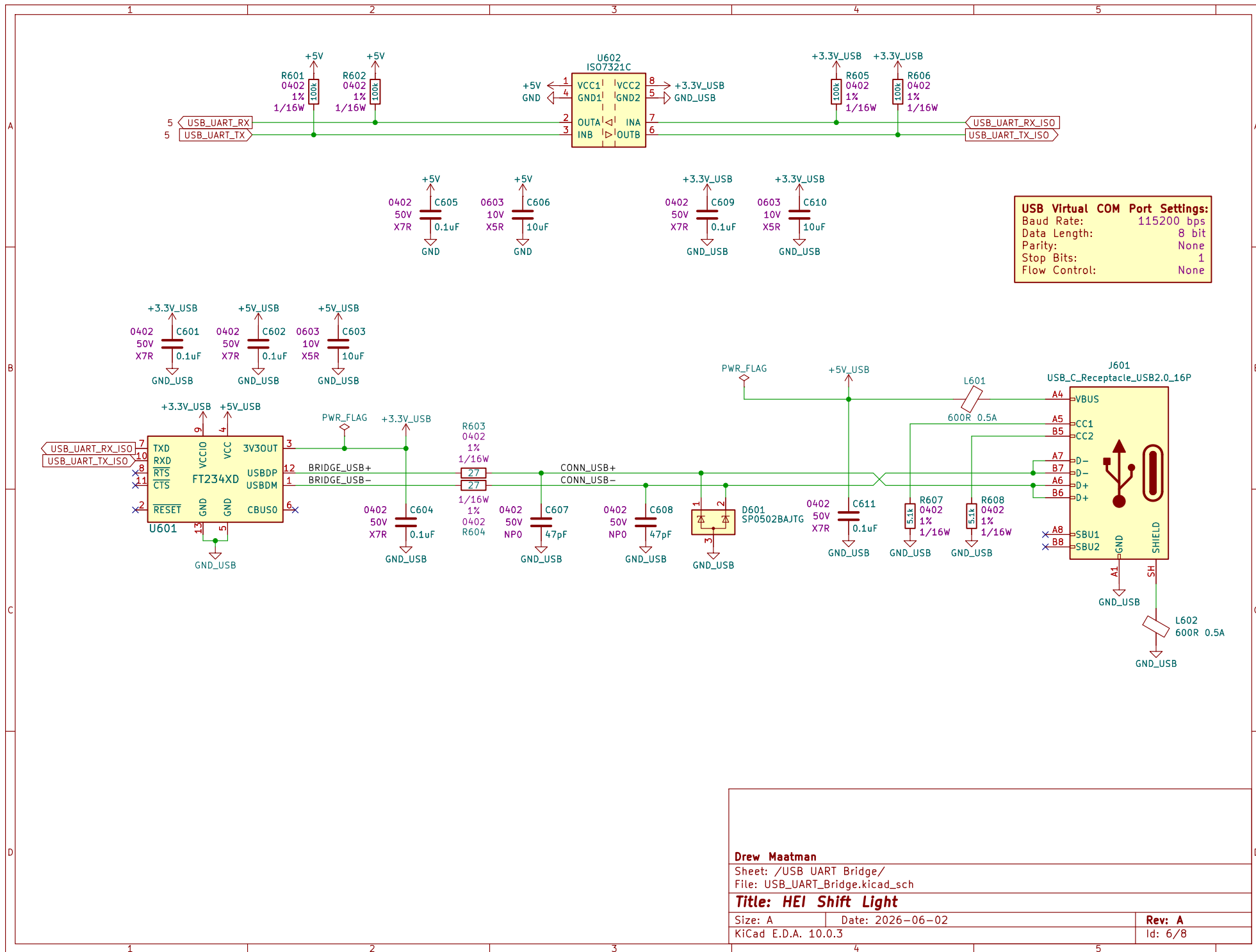
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Sheet: /Microcontroller/
File: microcontroller.kicad_sch

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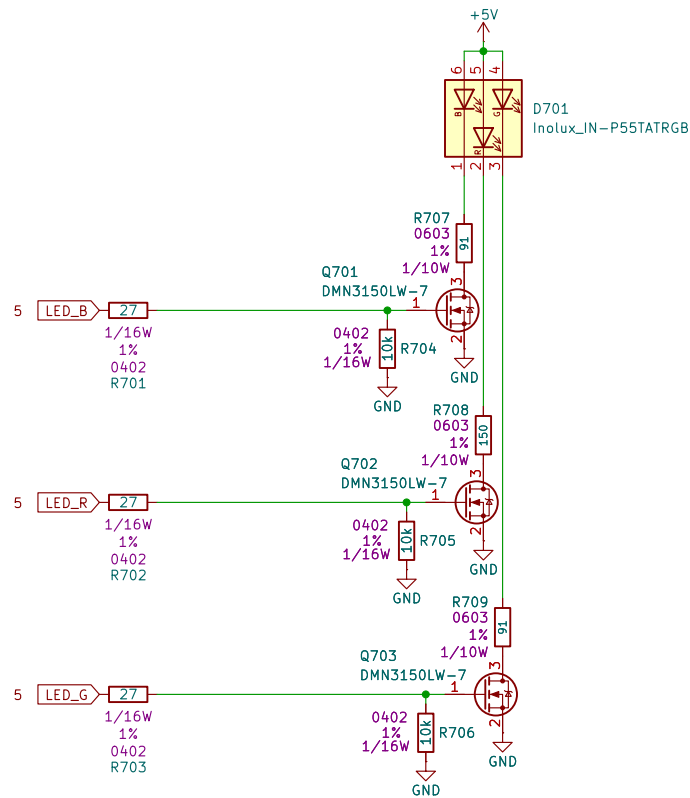
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Sheet: /USB UART Bridge/
File: USB_UART_Bridge.kicad_sch

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Based on the datasheet, the typical forward current (I_F) for testing is 20mA, with an absolute maximum of 25mA.
For "maximum brightness" while maintaining reliability, I recommend targeting 20mA.

Using Ohm's Law ($R = (V_{CC} - V_F) / I_F$ with $V_{CC} = 5V$:

Channel	Typ Vf	Calculation (20mA)	Recommended Res.	Max Bright (25mA)
Red	2.0V	(5 - 2.0) / 0.020	150 Ohm	120 Ohm
Green	3.2V	(5 - 3.2) / 0.020	91 Ohm (Standard)	72-75 Ohm
Blue	3.2V	(5 - 3.2) / 0.020	91 Ohm (Standard)	72-75 Ohm

Design Notes:

- Power Dissipation: At 20mA, the resistors dissipate approx 36-60mW. Standard 0603 or 0805 SMD resistors are more than sufficient.
- Safety: I strongly recommend the 20mA values. Pushing to 25mA provides a marginal increase in brightness but reduces the LED's lifespan and increases heat.

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Sheet: /RGB LED/
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